

GRANTS-MILAN, NM, USA

WMO: 723625

Lat: 35.165N

Lon: 107.902W

Elev: 1987

StdP: 79.62

Time Zone: -7.00 (NAM)

Period: 01-19

WBAN: 93057

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.9	-13.1	-23.0	0.6	-3.5	-20.9	0.7	-3.0	11.8	6.1	10.8	5.2	2.7	310	0.598

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.9	33.1	14.1	32.0	13.5	30.8	13.2	16.9	25.3	16.3	24.7	15.7	24.2	4.8	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.9	13.5	18.2	14.1	12.8	17.9	13.3	12.2	17.8	55.1	24.9	53.1	24.5	51.2	24.2	21.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.3	9.7	8.5	-20.8	35.5	3.8	1.2	-23.6	36.3	-25.8	37.0	-27.9	37.7	-30.7	38.6		
(5)			-21.2	18.2	3.5	0.6	-23.7	18.6	-25.8	18.9	-27.8	19.2	-30.4	19.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	-0.7	1.4	5.4	9.2	13.6	19.8	22.0	20.3	16.5	10.5	4.3	-0.3		
	DBStd	8.63	4.17	3.90	3.64	3.51	3.67	2.76	1.85	1.75	2.76	3.50	4.02	4.65		
	HDD10.0	1313	330	239	146	54	12	0	0	0	1	36	174	321		
	HDD18.3	3213	588	473	400	273	149	17	1	3	66	243	422	579		
	CDD10.0	1396	0	0	4	31	124	294	372	320	197	51	2	0		
	CDD18.3	253	0	0	0	0	3	60	114	65	11	0	0	0		
	CDH23.3	4809	0	0	2	17	319	1494	1516	1011	411	41	0	0		
	CDH26.7	1837	0	0	0	0	67	651	663	368	85	2	0	0		
(14) Wind	WSAvg	3.6	3.5	3.8	4.1	4.5	4.2	3.9	3.3	2.9	3.2	3.4	3.5	3.4		
(15) Precipitation	PrecAvg	248	10	9	10	15	13	13	42	55	24	21	14	15		
	PrecMax	398	45	39	54	64	69	43	101	123	73	160	67	67		
	PrecMin	140	0	0	0	0	0	1	5	24	0	0	0	0		
	PrecStd	66	12	10	14	17	16	12	24	25	18	32	15	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.3	19.1	23.1	25.2	30.9	34.7	35.0	33.1	30.4	26.4	21.4	16.6		
		MCWB	4.7	6.3	7.7	8.7	12.0	13.2	14.7	15.3	13.4	10.4	7.5	4.7		
	2%	DB	12.7	16.5	21.0	23.5	28.5	33.0	33.3	31.8	28.8	24.2	19.1	13.8		
		MCWB	3.8	5.0	6.9	8.1	10.6	13.0	14.5	14.4	12.9	9.8	6.8	4.2		
	5%	DB	10.8	13.8	18.9	22.2	26.9	32.0	32.1	30.4	27.5	22.7	17.1	11.3		
		MCWB	2.8	3.8	5.9	7.6	10.0	12.5	14.2	14.1	12.3	9.3	6.0	3.1		
	10%	DB	8.2	11.1	16.7	20.5	25.0	30.7	30.6	28.9	26.1	21.0	14.2	8.5		
		MCWB	1.7	2.9	5.1	6.8	9.2	11.9	14.2	14.3	12.1	8.9	4.9	2.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.7	7.2	8.3	10.3	13.6	16.3	18.0	18.0	16.1	13.9	8.7	6.7		
		MCDB	12.3	16.3	20.9	21.8	26.7	29.5	27.4	26.7	22.9	18.9	17.1	12.5		
	2%	WB	4.7	5.6	7.3	9.0	11.9	15.0	16.9	16.9	15.2	12.1	7.7	5.2		
		MCDB	10.5	14.1	19.4	20.8	24.4	26.6	25.7	24.7	21.7	19.0	16.2	11.7		
	5%	WB	3.3	4.5	6.4	8.0	10.8	14.0	16.4	16.4	14.5	10.8	6.7	3.9		
		MCDB	9.0	12.4	17.3	20.1	22.8	26.6	24.9	24.4	21.5	18.8	15.4	9.9		
	10%	WB	2.2	3.3	5.4	7.2	9.9	13.1	15.8	15.8	13.8	9.8	5.6	2.6		
		MCDB	7.4	10.2	15.5	19.1	22.5	26.3	24.6	23.7	21.3	18.5	13.7	8.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.4	16.9	19.2	19.1	19.6	20.6	16.9	17.0	17.8	18.8	18.5	16.0		
		MCDBR	20.5	23.0	23.9	22.5	22.3	22.4	19.8	20.4	20.7	22.2	22.5	20.1		
		MCWBR	12.5	13.0	12.4	11.3	10.2	9.4	6.7	7.3	8.1	10.5	12.3	12.0		
	5% WB	MCDBR	17.1	19.8	21.6	20.0	19.6	18.6	15.3	15.1	14.7	17.3	19.6	17.6		
		MCWBR	10.7	11.3	11.4	10.2	9.2	7.6	5.0	5.3	5.7	8.6	11.1	10.8		
(40) Clear-Sky Solar Irradiance	taub		0.233	0.238	0.260	0.279	0.291	0.310	0.354	0.333	0.296	0.270	0.244	0.235		
	taud		2.658	2.655	2.580	2.509	2.524	2.451	2.424	2.448	2.564	2.603	2.631	2.652		
	Ebn at Noon		1006	1031	1022	1007	991	966	921	938	967	975	981	984		
	Edh at Noon		69	78	92	105	105	113	115	110	93	81	70	66		
(44) All-Sky Solar Radiation	RadAvg		3.22	4.17	5.61	6.86	7.66	8.01	6.94	6.35	5.70	4.68	3.58	2.83		
	RadStd		0.27	0.34	0.35	0.36	0.49	0.41	0.36	0.26	0.22	0.36	0.22	0.18		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+33

Nomenclature: See separate page